

* Gradient Boosting Algorithm:

M-84

- i. Regression
- ii. Classification

Regression Data set

Exp	Degree	Salary	$\hat{y}$	$(y - \hat{y})$ R_1	R_2	$\hat{y}$	R_3
2	B.E	50K	75K	-25K	-28	72.7	-22.7
3	Masters	70K	75K	-5K	-8	74.7	-4.7
5	Masters	80K	75K	5K	3	—	—
6	PHD	100K	75K	25K	20	—	—
		Arg 75K					

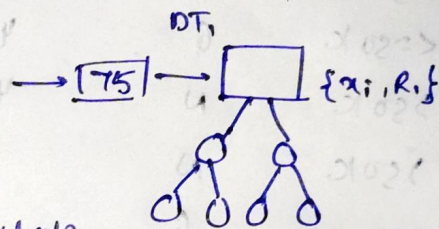
Steps

- i. Create a Base Model

$$\boxed{75} \quad \text{Average} = \frac{50 + 70 + 80 + 100}{4} = 75$$

- ii. Compute Residuals, Error

- iii. Construct a Decision Tree
consider input x_i and O/P R_i



Predicted O/P
 $= 75 + (-28) = 52$
 (overfitting)

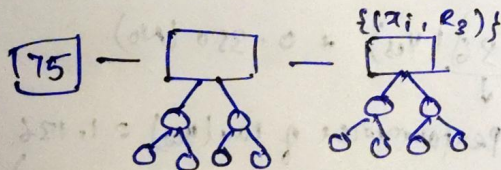
Predicted O/P $\Rightarrow 75 + \alpha(DT_1)$

$\alpha = \text{learning rate} \Rightarrow 75 + (0.1)(-28)$

$\alpha = 0.1 \quad (10\%) \Rightarrow 75 - 2.8 = 72.7$

$\Rightarrow 75 + 0.1(-8)$

$= 75 - 0.8 = 74.7$



Learning rate $\alpha = 0.1$

$$F(x) = \sum_{i=0}^n \alpha_i h_i(x)$$

Final function of Gradient Boosting.

$$F(x) = \alpha_0 h_0(x) + \alpha_1(h_1(x)) + \alpha_2(h_2(x)) + \alpha_3(h_3(x)) + \dots + \alpha_n(h_n(x))$$